

Teaching Computers To RECOGNISE CATS

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This article introduces the concept of machine learning. To get a sense of how it works, a simple example showing how a computer can recognise a cat from other animals is used. Python programming is used for implementing the same.

Machine learning is a rapidly growing field that enables computers to learn patterns found in the objects surrounding us. This may include tasks such as detection of an object from an image, speech recognition, autonomous driving and much more.

A typical machine learning follows the following steps:

Data collection. Like human beings, computers must be trained to distinguish between two types of animals—such as dogs and cats—by presenting these with a series of images. This is referred to as a training set of data. Fig. 1 shows one such training set, which contains a few images of cats and dogs. The larger and more diverse the training set, the better a computer can perform a learning task.

Features designing. Think about how you are able to tell the difference

between images of cats and dogs. What do you look for? You use colour, size, shape of ears or nose, and/or a combination of all to distinguish between the two animals.

You do not look at an image as only a collection of pixels. Instead, you pick out features from the image to identify either animal. This is true for computers as well. To successfully train a computer to perform the task of classification, you need to provide it with properly designed features.

This is not an easy task, because designing good quality features is application dependent.

For example, a feature like number of legs would be unhelpful in distinguishing between cats and dogs, as both have four legs. But, it would be useful in discriminating between cats and snakes.

Extracting features from an image can be challenging at times. For instance, if an image is blurred, features will not be properly extracted.



Fig. 1: Training set comprising cats and dogs

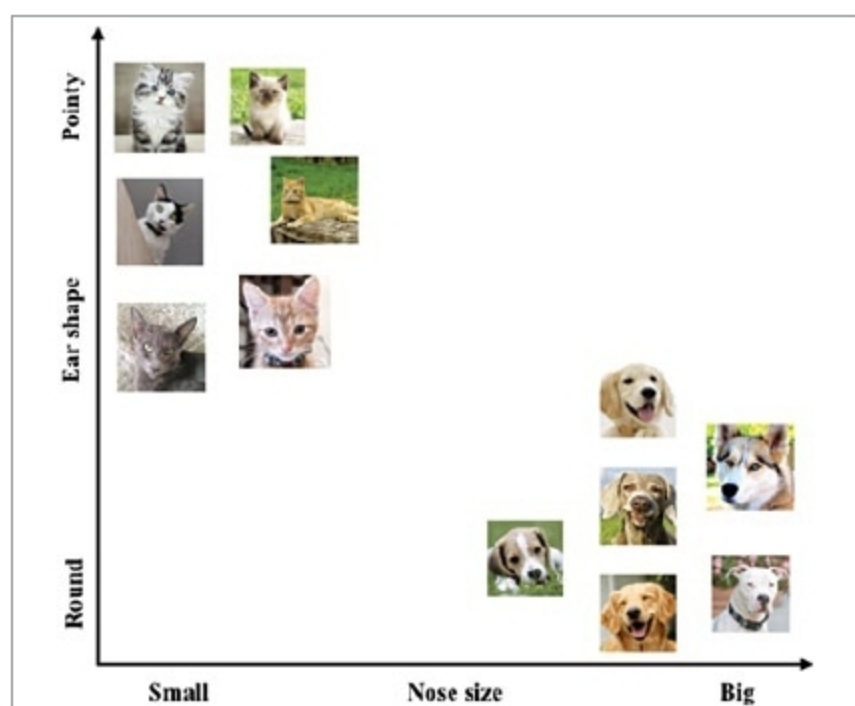


Fig. 2: Representation of training set in terms of features

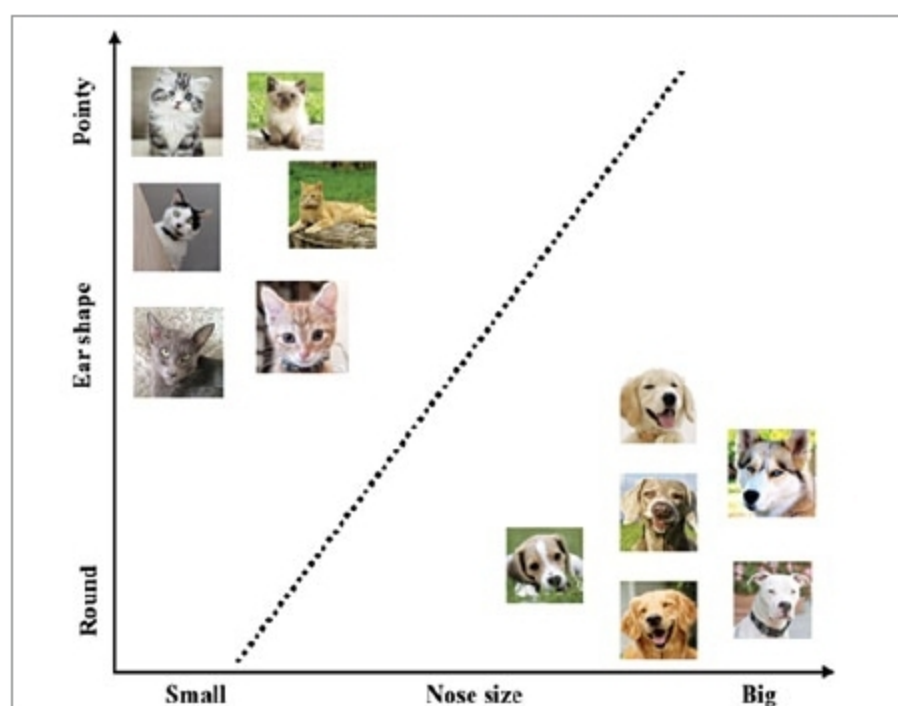


Fig. 3: Trained linear model (dotted lines) separating two classes of animals